UNIVERSIDAD ADOLFO IBÁÑEZ
Facultad de Artes Liberales
Departamento de Ciencias

CURSO: CORE CIENCIAS

El Artículo Científico:
Estructura IMRyD y Redacción de Artículos Científicos

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¿Qué es un artículo científico?

Es una forma de **comunicar** los resultados de una **investigación**, propuestas u opiniones, de forma **sistemática, clara**, **precisa**, y fundada en evidencia.

Un artículo científico debe:

-Poseer una estructura acorde (e.g., IMRyD).

-Estar enfocado en comunicar o debatir sobre nuevos conocimientos.

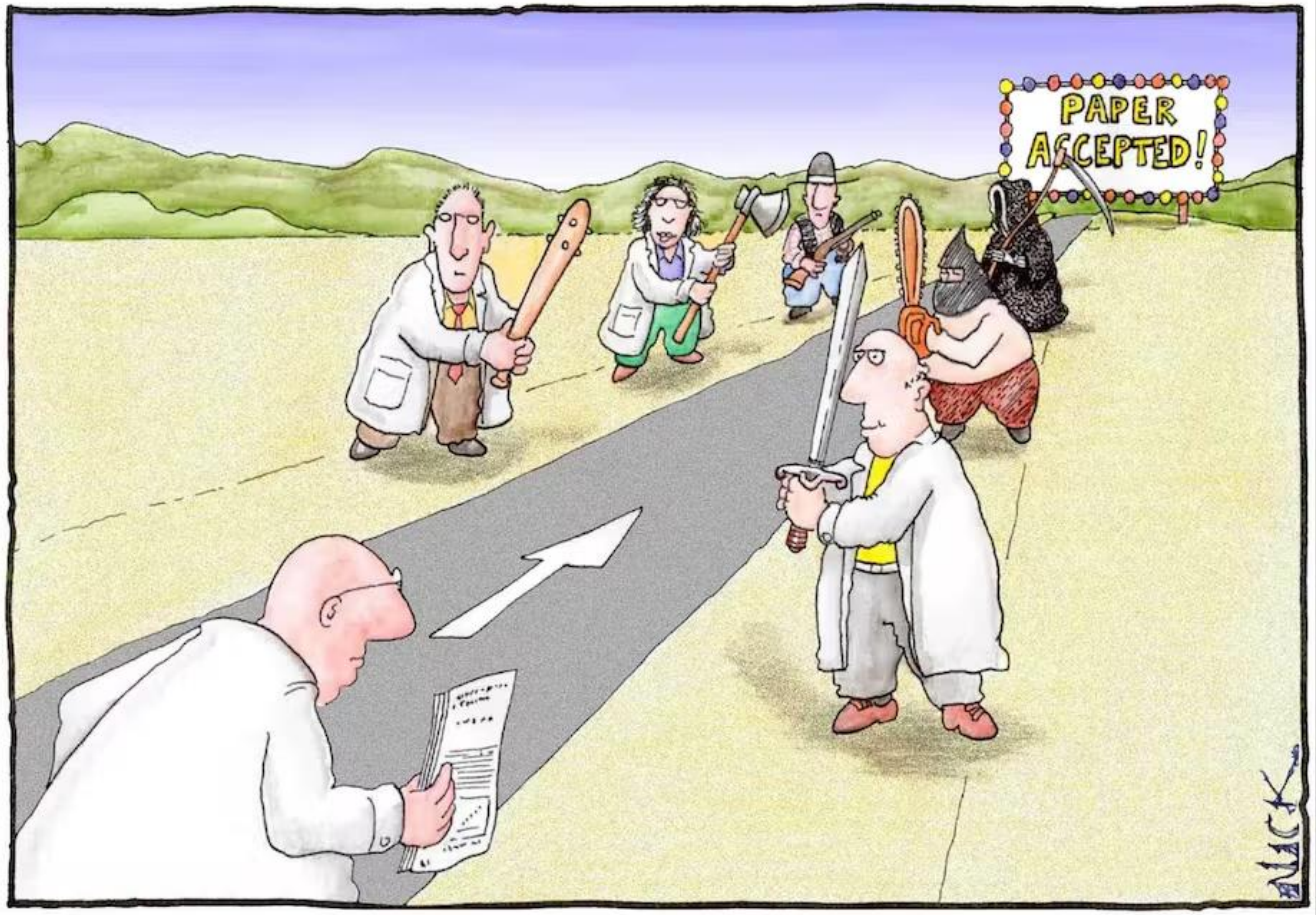
-Poseer **una sólida base de antecedentes** sobre el tema que aborda.

-Presentar **toda la información necesaria** para replicar el estudio o soportar los argumentos.

-Ser objetivo y presentar interpretaciones basadas en evidencias.

-Haber pasado por un proceso de revisión por pares.

(Las tesis no son consideradas artículos científicos!!!)



Most scientists regarded the new streamlined peer-review process as "quite an improvement."

Tipos de **artículos científicos**

Editoriales

- Cartas
- Notas
- Comentarios

Investigación

- Analíticos
- Casos estudio
- Propuestas
- Reportes

Revisiones

- Literatura
- Exploratoria
- Sistemática
- Meta Análisis

frontiers | Frontiers in Sustainable Cities

TYPE Editorial
PUBLISHED 29 November 2022
DOI 10.3389/frsc.2022.1101406

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OPEN ACCESS

EDITED AND REVIEWED BY
Stephan Pauleit,
Technical University of
Munich, Germany

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SPECIALTY SECTION

Editorial: Urban greening for ecosystem services provision: A Latin-American outlook

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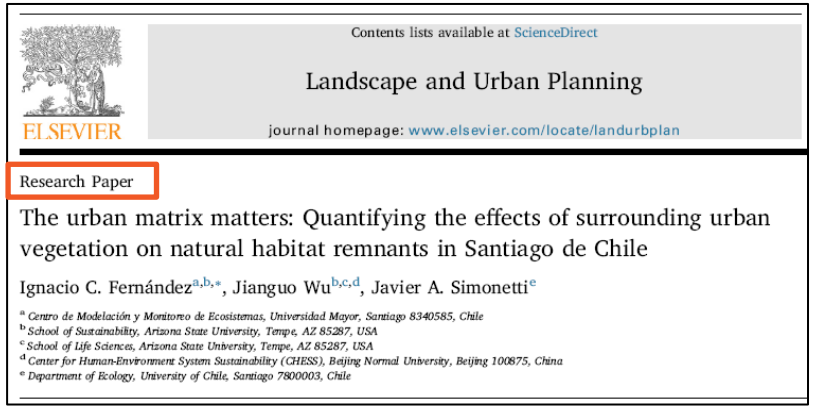
Contents lists available at ScienceDirect

Landscape and Urban Planning

journal homepage: www.elsevier.com/locate/landurbplan

ELSEVIER

Research Paper

The urban matrix matters: Quantifying the effects of surrounding urban vegetation on natural habitat remnants in Santiago de Chile

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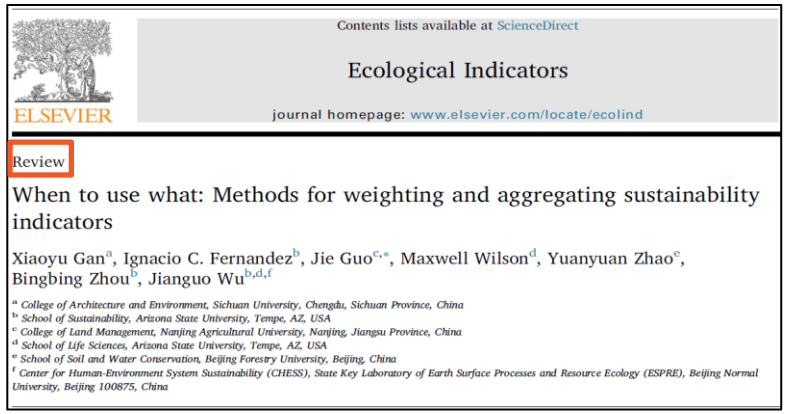
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Ecological Indicators

journal homepage: www.elsevier.com/locate/ecolind

ELSEVIER

Review

When to use what: Methods for weighting and aggregating sustainability indicators

Xiaoyu Gan[a], Ignacio C. Fernandez[b], Jie Guo[c,*], Maxwell Wilson[d], Yuanyuan Zhao[e], Bingbing Zhou[b], Jianguo Wu[b,d,f]

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¿Qué es la estructura IMRyD?

Es la estructura usada en la mayoría de las revistas de las áreas científicas, principalmente en ciencias naturales, formales y aplicadas.

El acrónimo **IMRyD** proviene de:

-**I**ntroducción
¿Cuál es el problema o pregunta a responder?

-**M**etodología
¿Cómo se abordó el problema o pregunta?

-**R**esultados
¿Qué se encontró al aplicar la metodología?

-**D**iscusión
¿Qué significan los resultados y cuál es su relevancia en función de lo que ya se sabe?

Anatomía de un artículo Científico

¿Son todas las Manzanas Rojas?

Resumen:

Analizamos el color de varias manzanas. Aunque la mayoría son rojas, algunas no lo son.

Introducción:

Una antigua pregunta es: ¿Son todas las manzanas rojas? Green (2004) indica que sí. Red (2006) señala que no. Este estudio busca resolver esta discrepancia.

Métodos:

Fuimos a 3 ferias y compramos todas las manzanas que tenían. Las miramos calmadamente y las clasificamos por color

Resultados:

Encontramos 200 manzanas rojas, 100 verdes y 50 amarillas.

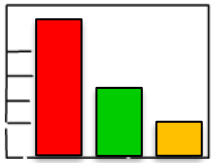

Figura 1

Discusión

Ya que encontramos manzanas verdes y amarillas, debe ser cierto que no todas las manzanas son rojas. Nuestros resultados apoyan lo planteado por Red (2006).

Referencias

- Green Ronald. (2004) *Ciencia y fruticultura.* 15(2): 118-129.
- Red Gerard. (2006) *Agricultura e innovación.* 12(3): 27-41

Modificado de "*How to Read a Scientific Paper*" de Michael Fosmire de la Universidad de Purdue

¿Qué debería contener cada sección?

Introducción:

- Principalmente texto. Rara vez se incorporan figuras, salvo cuando se requiera explicar un marco conceptual. Aquí se presenta el contexto de la investigación, sus antecedents, justificación, hipótesis y objetivos.

Metodología:

- Texto, diagramas y figuras/tablas. Se debe explicar cómo se hizo el estudio y entregar información suficiente para poder replicar el trabajo.

Resultados:

- Texto, figuras/tablas. Se presentan los resultados del estudio de la manera más gráfica posible, sin ningún tipo de interpretación más allá de los resultados de los análisis cualitativos y/o cuantitativos.

Discusión:

- Principalmente texto. Se deben interpretar los resultados en función de lo previamente conocido, y reflexionar respecto a sus implicancias. La conclusión es opcional, y suele utilizarse como un espacio para entregar un “mensaje refuerzo” al concluir el artículo.

¿Qué otra información contiene un artículo?

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Afiliación Autores

Editorial

World Development Sustainability 2 (2023) 100055

Revista

Título

Autores

Resumen

Keywords:
Environmental justice
Environmental inequity
Urban areas
Global South
Latin America

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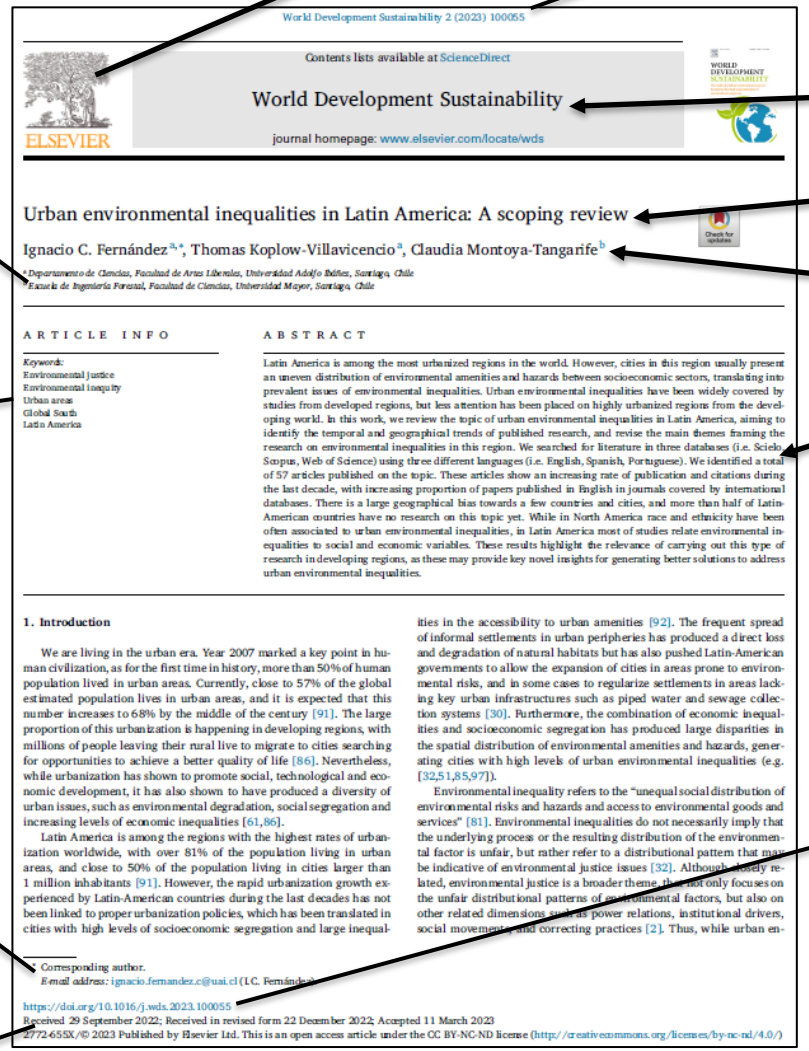
https://doi.org/10.1016/j.wds.2023.100055

Proceso de revisión por pares

Received 29 September 2022; Received in revised form 22 December 2022; Accepted 11 March 2023



World Development Sustainability 2 (2023) 100055

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World Development Sustainability

journal homepage: www.elsevier.com/locate/wds

Urban environmental inequalities in Latin America: A scoping review

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ARTICLE INFO

Keywords:
Environmental justice
Environmental inequity
Urban areas
Global South
Latin America

ABSTRACT

Latin America is among the most urbanized regions in the world. However, cities in this region usually present an uneven distribution of environmental amenities and hazards between socioeconomic sectors, translating into prevalent issues of environmental inequalities. Urban environmental inequalities have been widely covered by studies from developed regions, but less attention has been placed on highly urbanized regions from the developing world. In this work, we review the topic of urban environmental inequalities in Latin America, aiming to identify the temporal and geographical trends of published research, and revise the main themes framing the research on environmental inequalities in this region. We searched for literature in three databases (i.e. Scielo, Scopus, Web of Science) using three different languages (i.e. English, Spanish, Portuguese). We identified a total of 57 articles published on the topic. These articles show an increasing rate of publication and citations during the last decade, with increasing proportion of papers published in English in journals covered by international databases. There is a large geographical bias towards a few countries and cities, and more than half of Latin-American countries have no research on this topic yet. While in North America race and ethnicity have been often associated to urban environmental inequalities, in Latin America most of studies relate environmental inequalities to social and economic variables. These results highlight the relevance of carrying out this type of research in developing regions, as these may provide key novel insights for generating better solutions to address urban environmental inequalities.

1. Introduction

We are living in the urban era. Year 2007 marked a key point in human civilization, as for the first time in history, more than 50% of human population lived in urban areas. Currently, close to 57% of the global estimated population lives in urban areas, and it is expected that this number increases to 68% by the middle of the century [91]. The large proportion of this urbanization is happening in developing regions, with millions of people leaving their rural live to migrate to cities searching for opportunities to achieve a better quality of life [86]. Nevertheless, while urbanization has shown to promote social, technological and economic development, it has also shown to have produced a diversity of urban issues, such as environmental degradation, social segregation and increasing levels of economic inequalities [61,86].

Latin America is among the regions with the highest rates of urbanization worldwide, with over 81% of the population living in urban areas, and close to 50% of the population living in cities larger than 1 million inhabitants [91]. However, the rapid urbanization growth experienced by Latin-American countries during the last decades has not been linked to proper urbanization policies, which has been translated in cities with high levels of socioeconomic segregation and large inequalities in the accessibility to urban amenities [92]. The frequent spread of informal settlements in urban peripheries has produced a direct loss and degradation of natural habitats but has also pushed Latin-American governments to allow the expansion of cities in areas prone to environmental risks, and in some cases to regularize settlements in areas lacking key urban infrastructures such as piped water and sewage collection systems [30]. Furthermore, the combination of economic inequalities and socioeconomic segregation has produced large disparities in the spatial distribution of environmental amenities and hazards, generating cities with high levels of urban environmental inequalities (e.g. [32,51,85,97]).

Environmental inequality refers to the "unequal social distribution of environmental risks and hazards and access to environmental goods and services" [81]. Environmental inequalities do not necessarily imply that the underlying process or the resulting distribution of the environmental factor is unfair, but rather refer to a distributional pattern that may be indicative of environmental justice issues [32]. Although closely related, environmental justice is a broader theme, that not only focuses on the unfair distributional patterns of environmental factors, but also on other related dimensions such as power relations, institutional drivers, social movements and correcting practices [2]. Thus, while urban en-

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https://doi.org/10.1016/j.wds.2023.100055
Received 29 September 2022; Received in revised form 22 December 2022; Accepted 11 March 2023
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Las referencias

Ecological Economics 161 (2019) 193–201

Contents lists available at ScienceDirect

Ecological Economics

journal homepage: www.elsevier.com/locate/ecolecon

Methodological and Ideological Options

A multiple-class distance-decaying approach for mapping temperature reduction ecosystem services provided by urban vegetation in Santiago de Chile

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ARTICLE INFO

Keywords:
Urban sustainability
Urban vegetation
Services providing areas
Services benefiting areas
Ecosystem services mapping
Temperature regulation

ABSTRACT

Mapping ecosystem services is a challenging task, particularly for regulating services presenting spatial mismatches between the areas providing the services and those benefiting from them, such as the temperature reduction ecosystem service (TRES) provided by urban vegetation. The challenge for mapping TRES arises because vegetation not only reduces temperatures at the immediate location, but also in distance-decaying rates in surrounding areas. Furthermore, different functional types of vegetation have differential cooling capabilities, making difficult to assess the accumulated level of TRES provided by vegetated areas if these are heterogeneous in terms of vegetation cover and functional types. To help overcoming these challenges, this work presents a GIS-based methodological approach for mapping TRES that takes explicit consideration of the differential cooling capabilities of vegetation functional types and the distance-decaying spatial transference of services from vegetated to surrounding areas. This approach is applied to the city of Santiago, comparing the mapping results with two other potential alternative mapping methods. The mapping approach presented in this work consistently outperformed the alternative methods. This approach could be easily applied to other urban areas and adapted to be used for mapping other regulating services, such as air and noise pollution mitigation provided by urban vegetation.

1. Introduction

Urban areas are not decoupled from natural systems. On the contrary, they strongly depend on the ecosystem services provided by natural and semi-natural areas that are within and beyond the urban physical boundary (Grimm et al., 2008; Wu, 2013). Ecosystem services are defined as "the benefits people obtain from ecosystems", which are often categorized into provisioning (e.g. food, water, timber), regulating (e.g. air filtration, microclimatic regulation, flood mitigation), cultural (e.g. recreation, aesthetic, spiritual) and supporting (e.g. photosynthesis, soil formation, nutrient cycling) services (Millenium Ecosystem Assessment, 2005).

The large proportion of ecosystem services provided in cities are "imported" from outside urban boundaries (Luck et al., 2001). However, there is growing evidence that vegetated areas within cities play an essential role for providing several ecosystem services that are key for increasing urban quality of life (Mullaney et al., 2015; Tzoulas et al., 2007; Willis and Petrokofsky, 2017). For example, urban vegetation can reduce air pollution by influencing pollutants deposition and dispersion (Janhäll, 2015). Vegetation barriers located at the street level can decrease road traffic noise percolating to residential areas (Ow and Ghosh, 2017). Urban green spaces promote physical activities, improving people health and psychological well-being (Wolch et al., 2014). Urban vegetation could also decrease air and surface temperatures by evapotranspiration and shading effects, making it highly useful as a strategy for mitigating the adverse effects of the urban heat island effect (Gago et al., 2013). Therefore, planning urban vegetation for providing ecosystem services is becoming a key strategy for improving urban sustainability (Lovell and Taylor, 2013; Tzoulas et al., 2007).

Planning urban vegetation to increase ecosystem services requires to have a good understanding on the places where the provision of the required services is more necessary (Bodnaruk et al., 2017; Fernández and Wu, 2018; Locke et al., 2010; Morani et al., 2011). Consequently, mapping the current distribution of ecosystem services is essential for setting the baseline on where decision-makers can start planning interventions. Nevertheless, mapping ecosystem services provided by urban vegetation is still a challenging task (Pulighe et al., 2016), particularly for services that benefits areas beyond the places where these are been provided. In these cases, there could be a spatial mismatch between the services providing areas (SPA) and services benefiting

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https://doi.org/10.1016/j.ecolecon.2019.03.029
Received 25 September 2018; Received in revised form 19 March 2019; Accepted 26 March 2019
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improve the results, highlighting the relevance of considering the spatial patterns of ecosystem services transference from SPA to SBA for mapping regulating services. However, the NDVIDD model was outperformed when vegetation functional groups were used (i.e. MCVDD), instead of NDVI. Thus, results from this work support the assumption made for structuring the MCVDD approach, which is that is not only relevant to consider the patterns of the spatial transferences of services from SPA to SBA, but also the different capabilities of vegetational functional groups for providing TRES.

Because the different mapping approaches were compared through regression analysis, the good performance observed for the MCVDD approach could be thought to be an artifact of technical issues due to spatial autocorrelation (Dormann et al., 2007). Nevertheless, residual autocorrelations for the different models showed that autocorrelation was not affecting these results. This is further supported by the fact that the SLM model, which is specifically designed to handle autocorrelation issues, also showed that the MCVDD provides the map presenting the strongest spatial relationship between summer temperatures and Santiago's urban vegetation.

5. Concluding remarks

The MCVDD provides a simple methodological framework that could help improving the accuracy and reliability of maps representing the provision of several regulating services provided by urban vegetation at different scales. While there is ample evidence regarding the role of vegetation for providing regulating services in urban areas, methods explicitly considering the patterns of the spatial transference of services from SPA, such as the MCVDD presented here, have been scarcely covered in the literature. Furthermore, this method discriminates among the relative contribution of different functional types of vegetation for providing regulating services, which may provide insightful information for decision-makers. For instance, based on results for this work, Santiago's planners should prioritize the use of evergreen trees for providing TRES, and not deciduous and irrigated grass, which has been the common practice in this city. While in this work the MCVDD approach was used only for mapping TRES, the same approach could be used for mapping other regulating services such as noise and air pollution mitigation. As shown by the MCVDD results, integrating the relative contribution of different vegetation types with the distance-decaying functions for the transference of services from SPA, could generate mapping results that outperform others commonly used methods, such as raw NDVI values. Nevertheless, while the MCVDD method could provide adequate results for informing decision-making at coarser scales (metropolitan to municipalities level), still needs to be further refined for producing accurate results at finer scales (neighborhood to street level) where local level factors (e.g. infrastructure shape and roughness, building materials, irrigation patterns) could largely influence local microclimatic conditions. In this regard, advancing our understanding on the spatial transference of ecosystem services from SPA to SBA at finer scales still is a fundamental task for improving our capacity to better mapping and assessing ecosystem services provided by urban vegetation.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ecolecon.2019.03.029.

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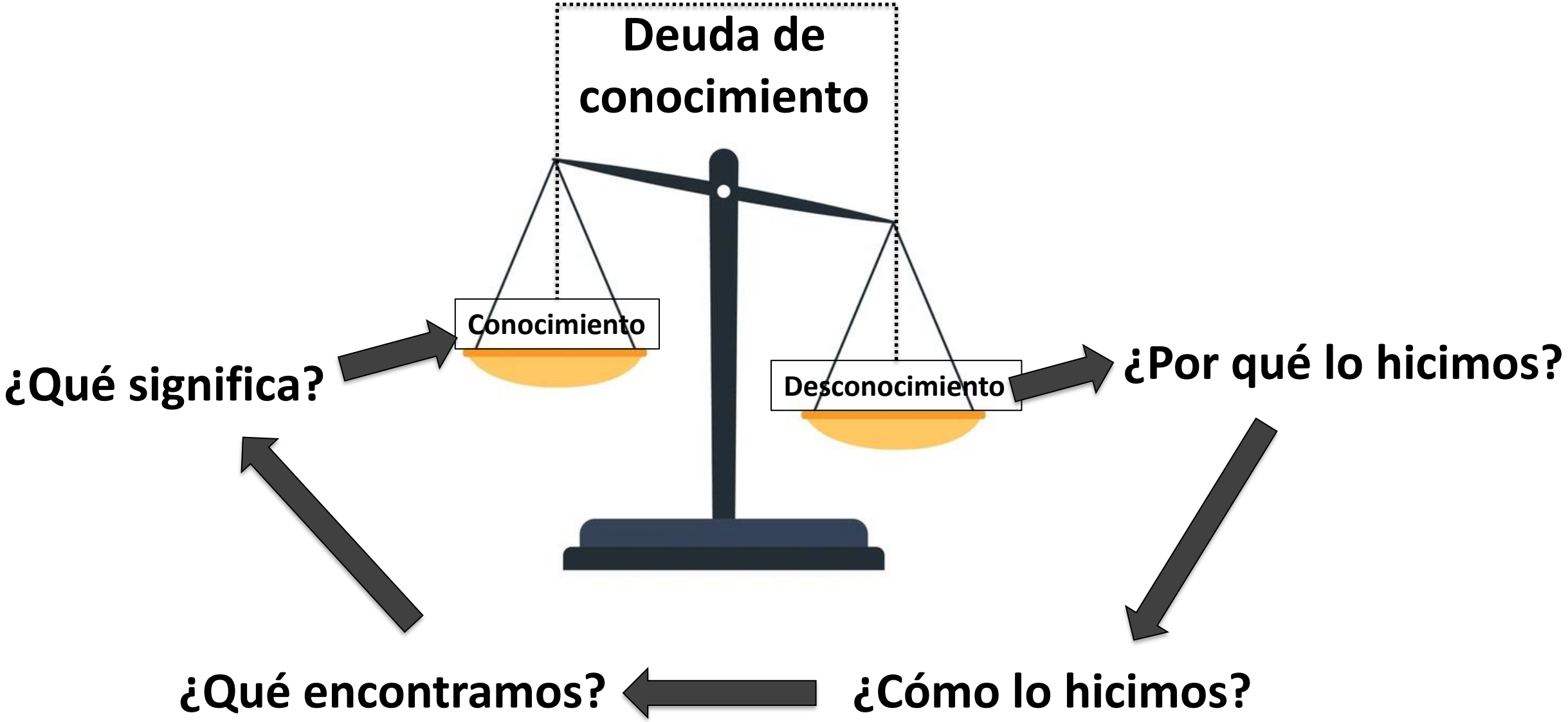
Las referencias son claves para:
-Sostener la introducción y relevar los objetivos.
-Justificar la metodología.
-Interpretar y poner en contexto los resultados.
-Facilitar el acceso a la información a los lectores.

Planning urban vegetation to increase ecosystem services requires to have a good understanding on the places where the provision of the required services is more necessary (Bodnaruk et al., 2017; Fernández and Wu, 2018; Locke et al., 2010; Morani et al., 2011). Consequently,

Bodnaruk, E.W., Kroll, C.N., Yang, Y., Hirabayashi, S., Nowak, D.J., Endreny, T.A., 2017. Where to plant urban trees? A spatially explicit methodology to explore ecosystem service tradeoffs. Landsc. Urban Plan. 157, 457–467. https://doi.org/10.1016/j.landurbplan.2016.08.016.

¿Cómo redactar un artículo científico?



¿Cómo redactar un artículo científico?

¿Por dónde comenzar?

Introducción
(Entrada)

Resultados
(Plato Principal)

Discusión
(Postre)

Metodología
(Receta y Preparación)

Resumen
(Menú)

¿Cómo redactar un artículo científico?

Tip: partir por el plato principal (resultados), pero teniendo muy claro cuál es la pregunta, hipótesis y objetivos del trabajo**.** El resumen es siempre lo último que se escribe.

Resultados
(Plato Principal)

Metodología
(Receta y Preparación)

Discusión
(Postre)

Introducción
(Entrada)

Resumen
(Menú)

Trabajo Grupal con Nota…pueden partir en clases

1) Genere un gráfico ficticio, como los ejemplos de abajo, que muestre algún resultado que responda una pregunta de investigación. El gráfico debe ser pertinente a su pregunta de investigación.
2) Desarrolle un resumen científico de entre 300 y 350 palabras, que incluya explícitamente las siguientes partes: Introducción, Metodología, Resultados, Discusión.
3) Incluya su gráfico y pregunta de investigación después del resumen. Subir en una plana en Webc.

Gráfico de Barras/Columnas

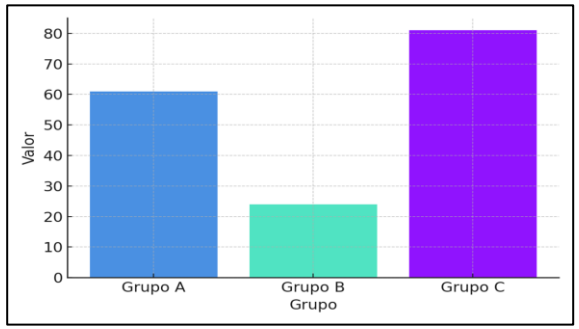


Gráfico de Dispersión

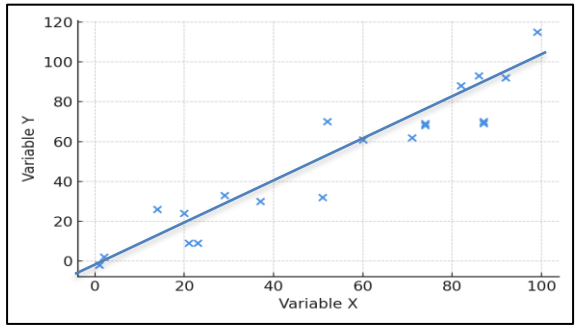


Gráfico de líneas

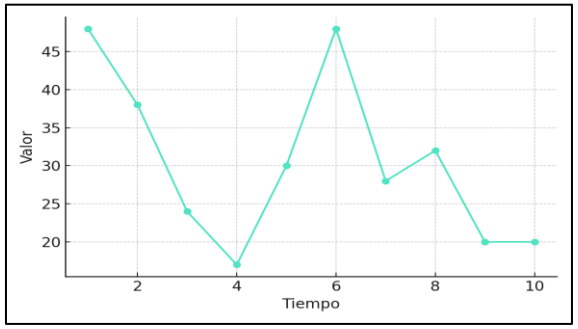


Gráfico Cajas (y Bigotes)

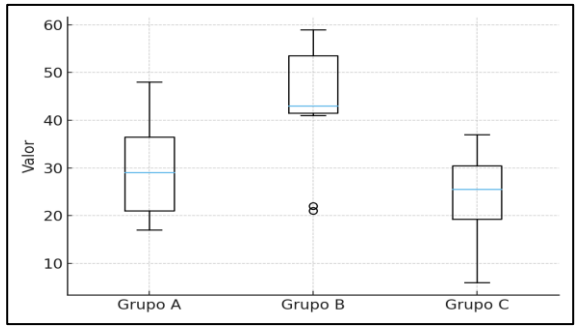
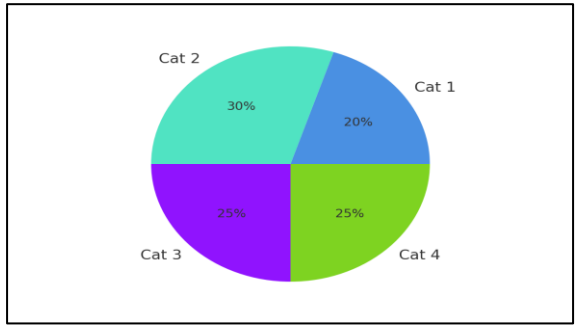
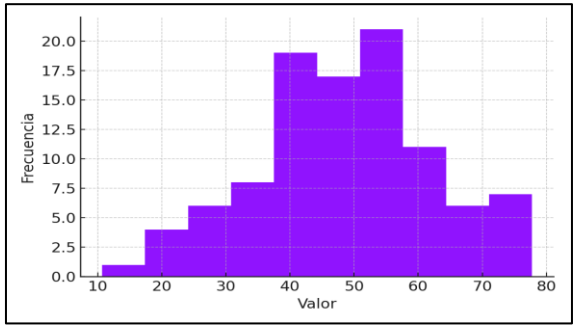


Gráfico de Torta



Histograma